

Figure 2: The internal structure of Kubernetes

the cluster, using the Kubernetes package manager called Helm.

Kubernetes internals

Kubernetes has been created by Google, based on its experience in running huge workloads. It is designed to work in different environments such as virtual machines, on bare metal systems or on the cloud. It is completely open source and is widely used these days. It is very modular and can be customised up to the core. It offers multiple options for setting up the cluster, right from networking to container runtimes.

There are some important components of Kubernetes that we need to understand before jumping into the installation process. Figure 2 shows a typical Kubernetes cluster, with master nodes and slaves (simply called nodes). The master node takes care of the slaves and the deployment of the various components of Kubernetes in each of them. The master node uses *etcd* to store the information about what is to be deployed at what point in time in each of the slaves, and hence maintains the state of the cluster. A cluster can be formed using more than one system, which is why the components are packaged into nodes.

Details of some other important components of a Kubernetes cluster are given below.

- **Pods:** A group of containers on the same host is called a pod, and all the containers in a pod share the same network settings such as the Internet Protocol (IP) address.
- **Cluster:** A cluster is a set of nodes whose resources are used to run a set of applications.
- **Services:** The services provided are load balancers for the pods in a cluster, exposing them to the outside world.
- **Replication controller:** This controls the number of replicas of each pod in a cluster, and also makes sure the number of pods is equal to the replica count set in the configuration.

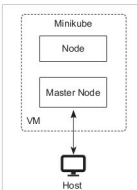


Figure 3: How Minikube works

- **Labels:** Labels are tags used to differentiate pods, services and other components.
- **Namespaces:** Namespaces help in managing pods in environments where developers are spread across multiple projects. They divide cluster resources between multiple users. Namespaces can be skipped in clusters with less than ten users.

Setting up a Kubernetes cluster using Minikube

Minikube, which is a part of the official Kubernetes project, is a command line tool to install and run Kubernetes clusters locally. The cluster can be started or stopped, just like any other service in Linux. Minikube runs all the components of the cluster like the nodes, pods, etc, in a virtual machine on the host. Therefore, it is required to either have VirtualBox or Kernel Virtual Machine (kvm) installed on the host system. Minikube, however, can only run a single node cluster.

Installing Minikube

Run the commands as shown in Figure 4.

```
$ curl -io minikube https://storage.googleapis.com/minikube/releases/v0.25.0/minikube
-linux-amd64 && chmod +x minikube && sudo mv minikube /usr/local/bin
% Total % Received % Xferd Average Speed Time Time Current
100 41.0M 100 41.0M 0 7324k 0 0:00:05 0:00:05 --:--:-- 9714k
$
[sudo] password for vleeet:
```

Figure 4: Minikube installation commands

The *curl* command first checks if the location is correct or needs to be redirected using the *L* flag. It then downloads the latest version of Minikube and saves the output to a file instead of *stdout*, using the *o* flag. The *&&* command is used to run multiple commands sequentially, one after the other but as a single command. The *chmod* command gives appropriate permissions to the downloaded file, and the last command moves it into the */usr/local/bin* folder so that it can run from any directory since the *PATH* is set by default.

How to talk to the cluster

Install Kubectl, which is a command line tool, to talk to the cluster and get the necessary information about it. It is like a medium of communication between the user and the cluster. Kubectl can be used to start, modify or delete parts of the cluster. More details on Kubectl can be found at <https://kubernetes.io/docs/setup/pick-right-solution/>. To install Kubectl, run the commands as shown in Figure 5.

Similar to the Minikube installation, these commands download the binary file of Kubectl, give it appropriate permissions and move it to */usr/local/bin*. The inner *curl* command extracts the latest version number.

```
$ curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/release/$(cat /etc/os-release | grep ID | cut -d= -f2) | tr -d '\n')/linux/amd64/kubectl
% Total % Received % Xferd Average Speed Time Time Current
100 51.7M 100 51.7M 0 7875k 0 0:00:06 0:00:06 --:--:-- 9.8M
$
```

Figure 5: Kubectl installation commands